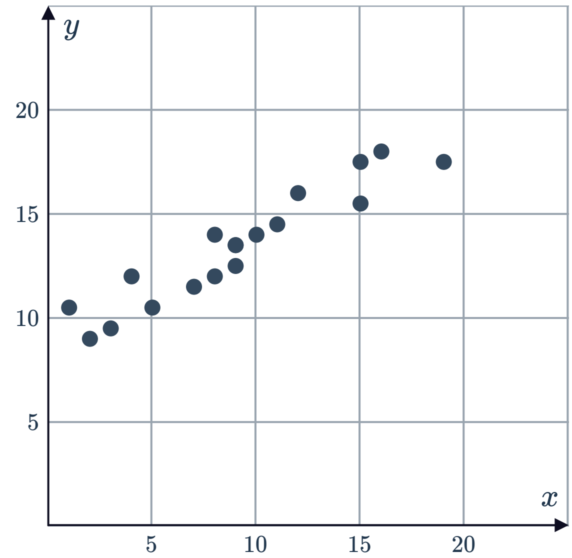
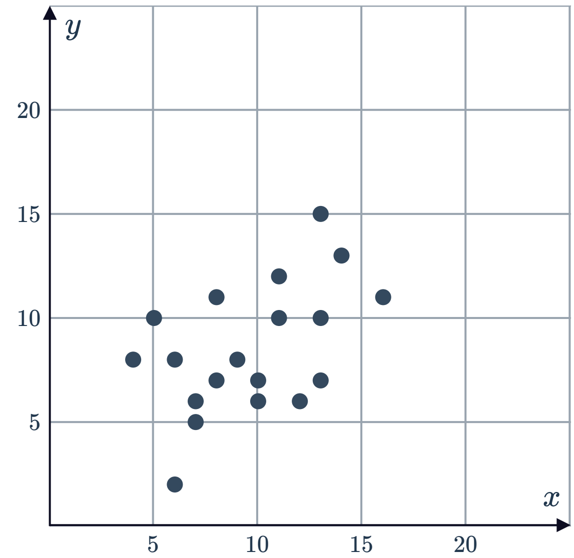
Pearson's correlation coefficient

1 Describe the type of linear relationship between the variables in the following scatter plots:

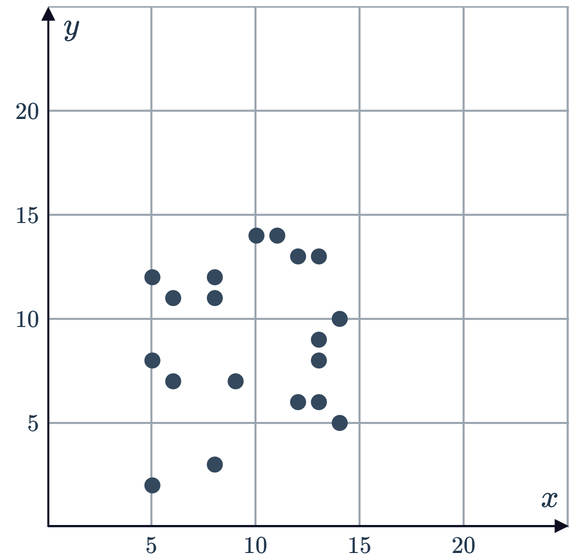
a



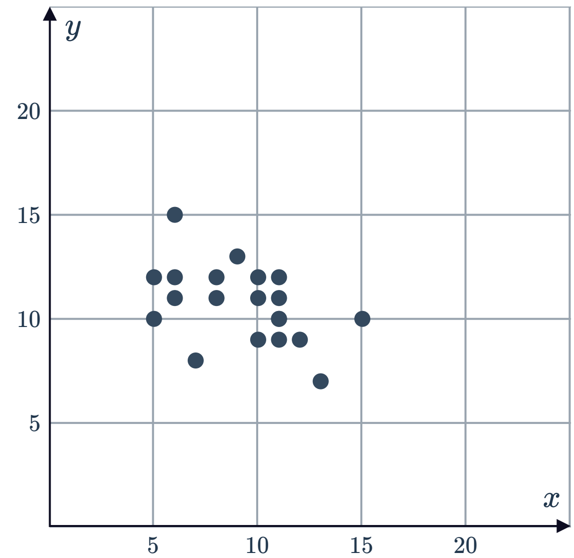
b



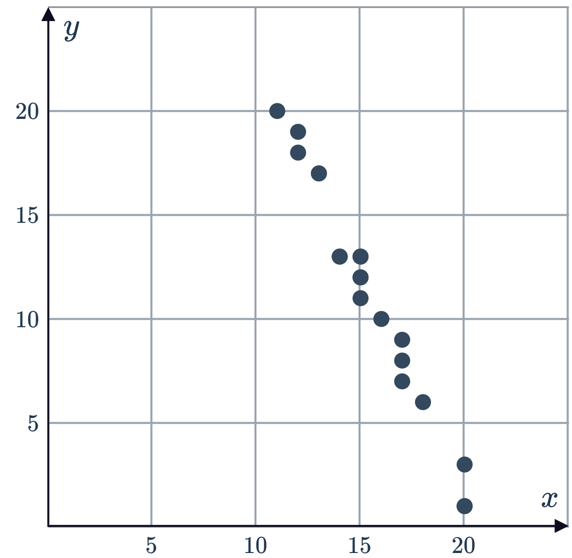
c



d

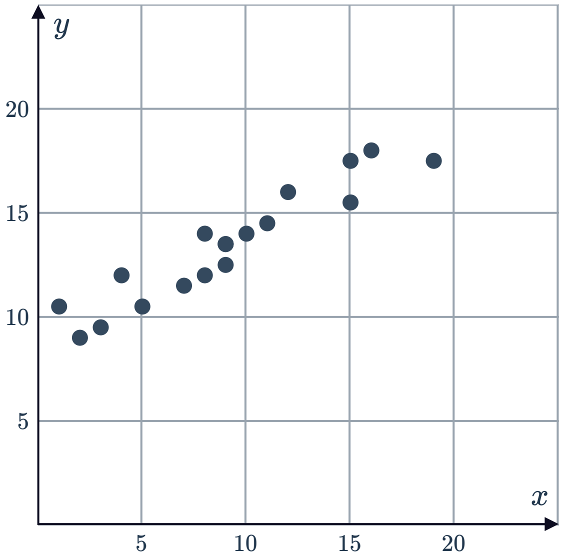


e

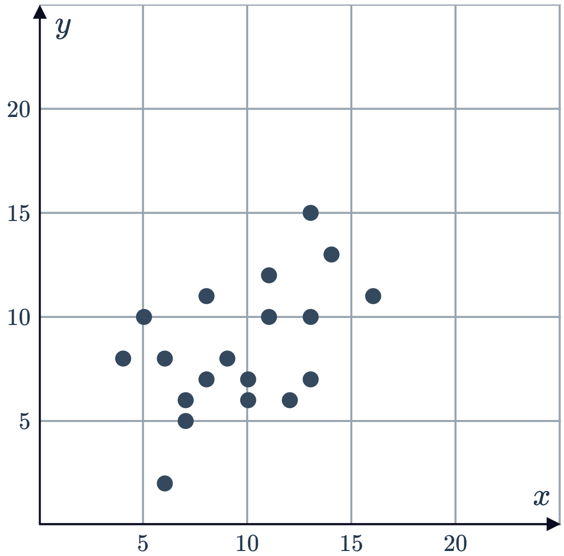


2 Estimate the value of the correlation coeff[redacted] for the data in the following scatter plots:

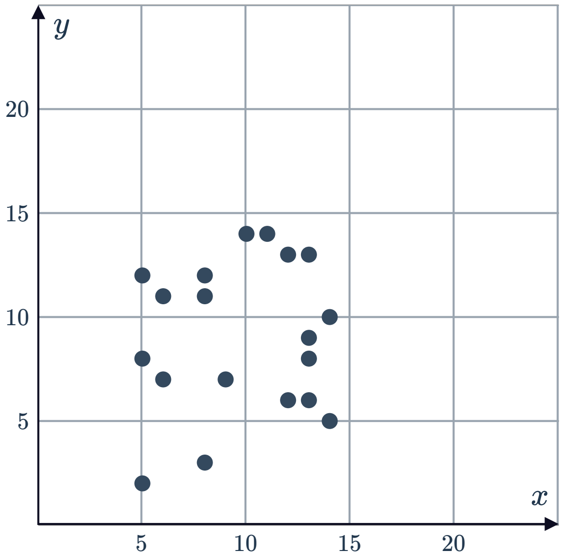
a



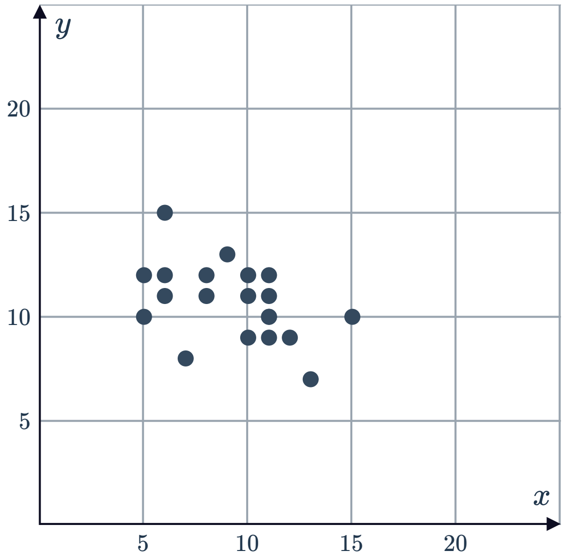
b



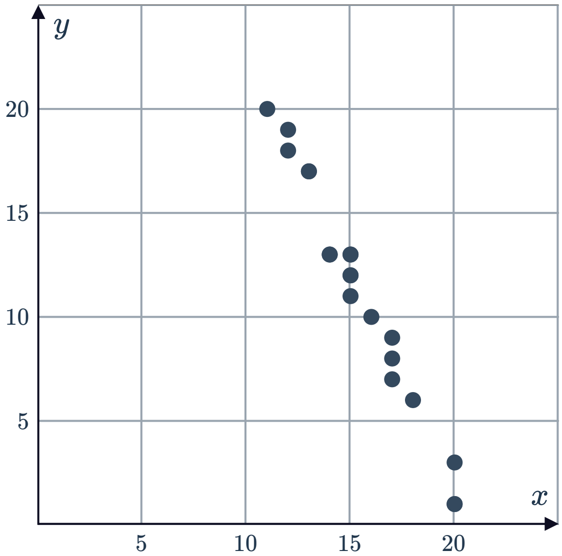
c



d



e



3 Describe the linear relationship between t[redacted]riables with the following correlation coefficients:

a 0.96 b 0.66 c 0.36 d -0.06

e -0.34 f -0.66

4 If the explanatory variable increases, describe the effect on the response variable for the following studies:

a A study found that the correlation coefficient between heights of women and probability of being turned down for a promotion was found to be -0.90.

b A study found that the correlation coefficient between population of a city and number of speeding fines recorded was found to be 0.83.

c A study found that the correlation coefficient between length of hair and length of fingernails was found to be 0.07.

d A study found that the correlation coefficient between number of bylaws a council has about dog breeding and number of dogs available for adoption at the local shelter was found to be 0.55.

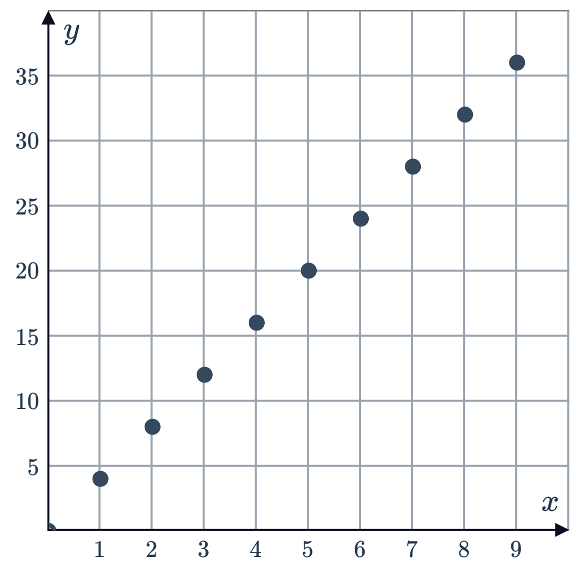
5 A pair of data sets have a correlation coefficient of $\dfrac{1}{10}$ while a second pair of data sets has a correlation coefficient of $\dfrac{3}{5}$. Which pair of data sets have the stronger correlation?

6 Describe the type of correlation the following correlation coefficients indicate:

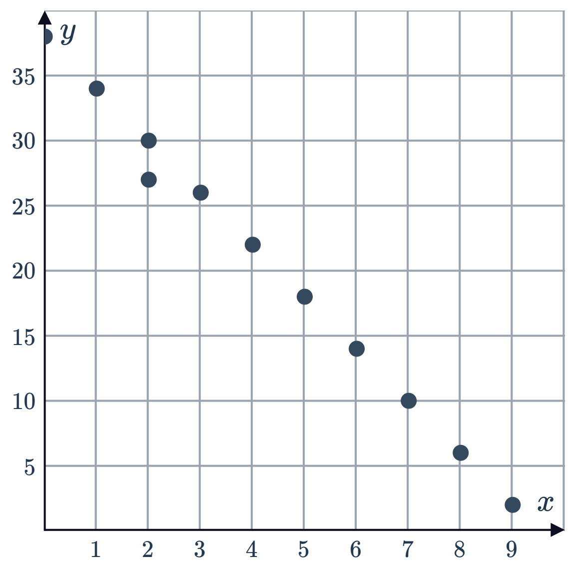
a $r = 1$ b $r = 0$ c $r = -1$

7 For each of the following graphs, write down an estimate of the correlation coefficient:

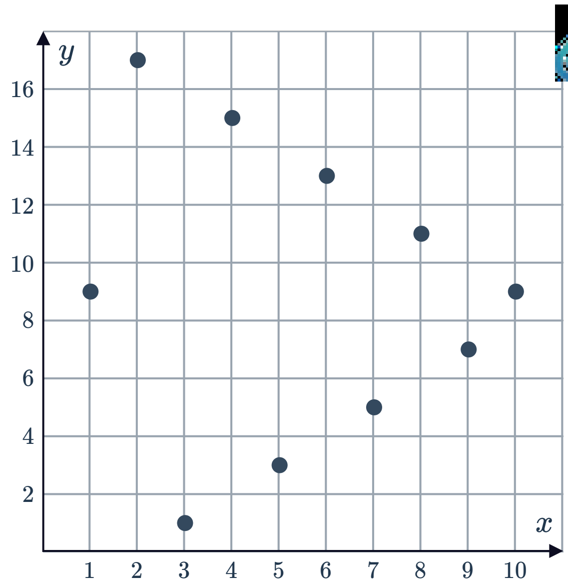
a



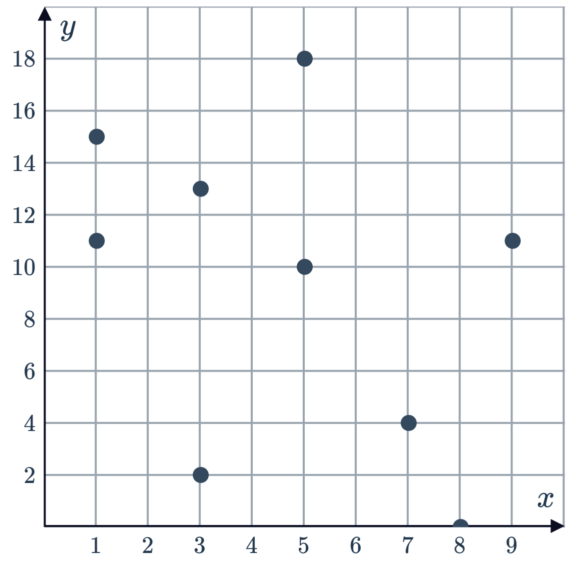
b



c



d



8 For each of the following pairs of relationships:

- **i** Determine whether the two linear relationships have the same direction.
- **ii** State the relationship that has the stronger correlation.

a The linear relationship between a set of data for variables x and y has a correlation coefficient of 0.3. The linear relationship between a set of data for variables x and z has a correlation coefficient of 0.9.

b The linear relationship between a set of data for variables x and y has a correlation coefficient of -0.9. The linear relationship between a set of data for variables y and z has a correlation coefficient of -0.5.

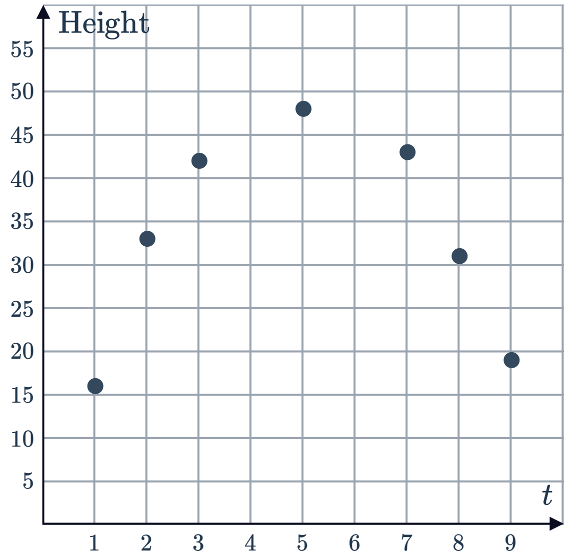
c The linear relationship between a set of data for variables x and y has a correlation coefficient of 0.8. The linear relationship between a set of data for variables s and t has a correlation coefficient of -0.5.

d The linear relationship between a set of data for variables x and y has a correlation coefficient of 0.3. The linear relationship between a set of data for variables x and t has a correlation coefficient of -0.9.

9 A researcher plotted the life expectancy of a group of men against the number of cigarettes they smoke a day. The results were recorded and the correlation coefficient r was found to be -0.88.
Describe the correlation between the life expectancy of a man and the number of cigarettes smoked per day.

10 A researcher was evaluating the relationsh[redacted]tween the number of years in education a person completes and the number of pets they own. The results were recorded and correlation coefficient r was found to be -0.3.
Describe the correlation between a person's years of education and the number of pets they own.

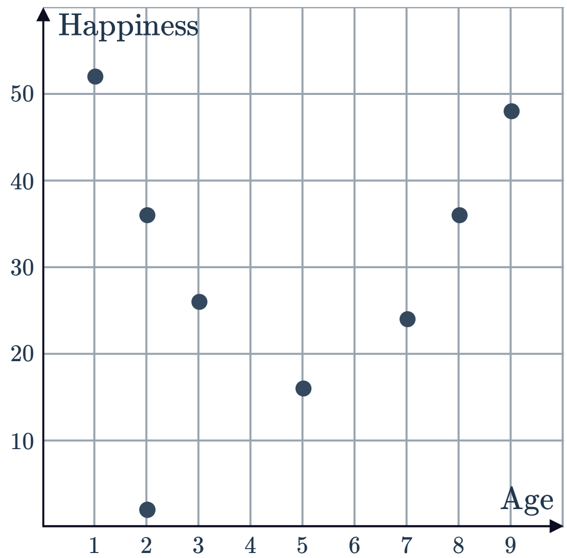
11 The scatter diagram shows data of the height of a ball kicked into the air as a function of time:



a Which type of model is appropriate for the data, linear or non-linear?

b Write down a possible value of Pearson’s correlation coefficient, r, for this set of data.

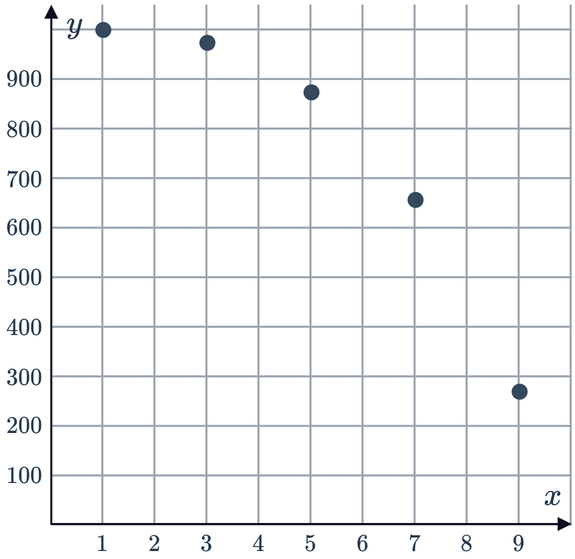
12 The scatter diagram shows data of a person's level of happiness as a function of their age:



a Which type of model is appropriate for the data, linear or non-linear?

b Write down a possible value of Pearson’s correlation coefficient, r, for this set of data.

13 The scatter diagram shows data of the height of an object after it is pushed off a rooftop as a function of time:



a Which type of model is appropriate for the data, linear or quadratic?

b Write down a possible value of Pearson's correlation coefficient, r, for this set of data.

Correlation vs causation

14 Determine whether the following describe a relationship that is correlated but not causal:

a The sales of ice cream and increase in temperature.

b The number of hours worked and how much money is made for a given person.

c The amount of showers had in a day and the amount of the water bill.

d The amount of rainfall received, and level of water in a lake.

e The larger the dimensions of a rectangular verandah, the more area.

f The season of the year and the number of water related injuries.

g Increase in temperature, and the level of mercury in a thermometer.

h The number of students shouting in class and the number of detentions received.

15 For each of the following data examples, determine if there is a causal relationship between the variables:

a The number of times a coin lands on heads and the likelihood that it lands on heads on the next flip.

b The amount of weight training a person does and their strength.

16 Determine whether the following describe a causal relationship and not just a correlation:

a An individual's decision to work in construction and his diagnosis of skin cancer.

b The number of minutes spent exercising and the amount of calories burned.

c A decrease in temperature and the increase in attendance at an ice skating rink.

d As a child's weight increases, so does her vocabulary.

17 Determine whether the following are examples of variables with no correlation:

- **a** The age of a child and their shoe size.
- **b** The age of a child and their height.
- **c** The age of a child and the number of pets owned.
- **d** The age of a child and the amount of adjectives learned.

18 The table shows the number of fans sold at a store during days of various temperatures:

Temperature ($^\circ$C)	6	8	10	12	14	16	18	20
Number of fans sold	12	13	14	17	18	19	21	23

- **a** For this data, will r be greater than zero, less than zero or equal to zero?
- **b** Is there a causal relationship between the variables? Explain your answer.

19 A study found a strong positive association between the temperature and the number of beach drownings.

- **a** Does this mean that the temperature causes people to drown? Explain your answer.
- **b** Is the strong correlation found a coincidence? Explain your answer.

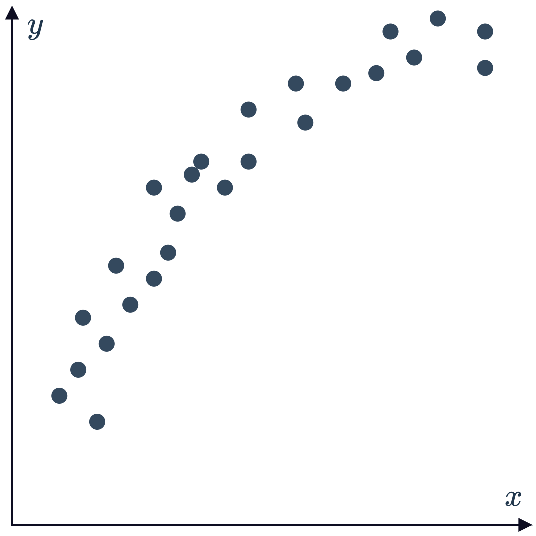
20 Many trees lose their leaves in winter. Does this mean that cold temperatures cause the leaves to fall?

21 A study found a strong correlation between the approximate number of pirates out at sea and the average world temperature.

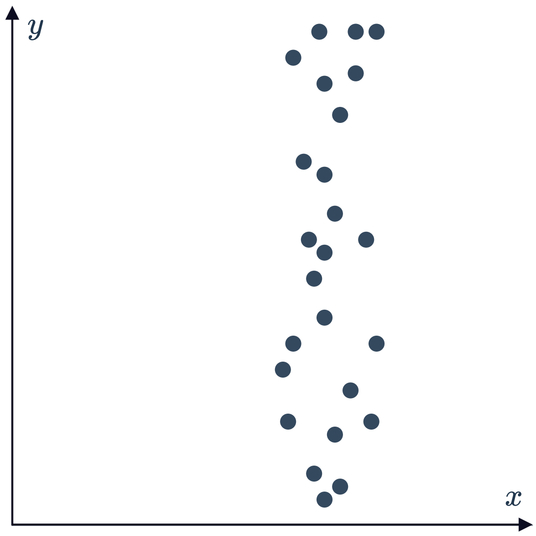
- **a** Does this mean that the number of pirates out at sea has an impact on world temperature?
- **b** Is the strong correlation found a coincidence? Explain your answer.
- **c** If there is correlation between two variables, is there causation?

22 Scatter plots for two sets of data are shown below:

Set A



Set B



a Which data set has the strongest linear correlation between the variables?

b Which data set appears to have a non-linear relationship? Explain your answer.

c Why does data set B have the weakest correlation between its variables?

d State whether the following pairs of variables could be represented by data set B:

- **i** Marks in an English examination and distance travelled from home to school.
- **ii** Cost of cars and cost of petrol.
- **iii** Distance travelled in a car and the cost of a driver's license.
- **iv** Height and weight of students at school.